

Daily Research Digest — 2026-04-21

Topics: galaxy clusters, galaxies, gravitational lensing, dark matter.

Synthesis

Executive overview

Recent arXiv preprints offer insights into galaxy formation and evolution through high-resolution spectroscopic and imaging studies. The research highlights the complex dynamics of gas flows in disk galaxies, the spatially resolved characteristics of molecular clouds at galactic edges, and the magnetic field properties of clusters like Fornax. Gravitational lensing is explored with a focus on higher-order statistics to understand non-Gaussianities in mass distributions. Dark matter studies include potential detections through astrometric precision improvements and neutrino observatory results probing dark matter annihilation.

Galaxy Clusters

- ****Magnetic Fields Characterization****: The MeerKAT Fornax Survey provides detailed magnetic field properties of the Fornax cluster, using a dense RM grid. Magnetic field power spectra are analyzed with a slope of ~ 2.7 fluctuating between ~ 1 and ~ 15 kpc, revealing central strengths of $\sim 5\mu\text{G}$ decreasing with thermal plasma density.

Galaxies

- ****Cold Gas Dynamics****: NOEMA3D studies reveal ordered molecular gas rotation with moderate turbulence in main-sequence galaxies at $z \sim 1.1$ to 1.6 , finding significant spatially coherent velocity residuals interpreted as radial flows.
- ****Molecular Clouds and Halos****: Spatially resolved studies of molecular clouds show varying star formation activities and color distributions across different cosmic web patterns. Velocity anisotropy profiles in the Milky Way's stellar halo are detailed up to ~ 70 kpc, indicating a transition from radially-dominated inner regions to tangentially-dominated outer regions.
- ****Magnetic Fields****: The Fornax clusters magnetic field is characterized using RM grids with high sensitivity and resolution.

Gravitational Lensing

- **Fourth-order Statistics**: Research extends the analytical framework for correlations between galaxies and shear to fourth order, focusing on aperture statistics. This work derives a four-point correlation function (4PCF) and its relation to the trispectrum, enhancing understanding of non-Gaussian features in mass distributions.

Dark Matter

- **Globular Cluster Halos**: GaiaNIR mission simulations predict improved astrometric precision for detecting dark matter halos in globular clusters like M4 under heavy extinction conditions.
- **IllustrisTNG Caustic Skeleton**: Analysis associates galaxies with voids, walls, filaments, and cluster nodes to discern their properties across the cosmic web. Blue galaxy fractions are studied as functions of cosmic web scales.

Neutrino Observations

- **IceCube Results**: Latest IceCube results include identification of steady neutrino sources like NGC 1068, measurements on flavor composition, limits on atmospheric neutrinos, and searches for neutrinos from dark matter annihilation in the Sun. Future upgrades aim to enhance sensitivity and expand detection capabilities.

Summary

The recent studies highlight advancements in understanding galaxy dynamics through detailed spectroscopic and imaging analyses of gas flows, magnetic fields, and spatially resolved properties. Gravitational lensing research progresses by incorporating higher-order statistical measures. Dark matter investigations involve precision astrometry and neutrino observatory results to probe the elusive nature of dark matter.

Future Work

Future studies should continue exploring higher-order gravitational lensing statistics for non-Gaussianities and extend high-resolution spectroscopic surveys for more comprehensive galaxy dynamics. Enhancements in astrometric precision will be crucial for detecting dark matter halos, while expanding neutrino observatory capabilities can further elucidate dark matter interactions and annihilation processes.

Paper Summaries and Figures

1. A Census of Na D-traced neutral ISM and outflows at $0.6 < z < 4$

Focus: galaxies

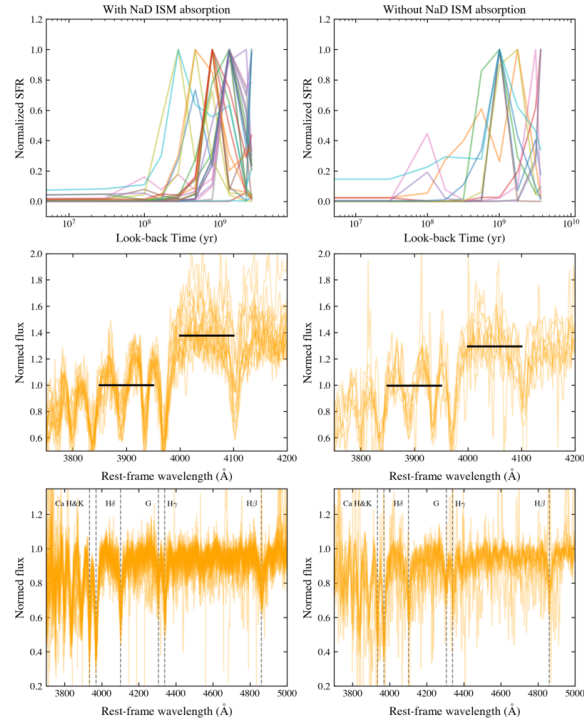
Authors: Yang Sun, Zhiyuan Ji, Francesco D'Eugenio, Yongda Zhu, George H. Rieke, William M. Baker, Andrew J. Bunker, Stefano Carniani, Jakob M. Helton, Michele Perna, Pablo G. Pérez-González, Pierluigi Rinaldi, Hannah Übler, Christopher N. A. Willmer

Published: 2026-04-20T17:19:44+00:00

PDF: <https://arxiv.org/pdf/2604.18522v1>

Summary: We present a statistical census of the Na D-traced neutral interstellar medium (ISM) and outflows in 309 galaxies at $0.6 < z < 4$ using JWST/NIRSpec medium-resolution grating spectroscopy from the SMILES, JADES, Blue Jay, and Aurora surveys. After subtracting the stellar continuum, we model the Na D $\lambda\lambda 5890, 5896$ Å and detect neutral ISM absorption in 76 galaxies. Of the Na D-traced ISM detections, 85\% are found in massive galaxies ($\log(M_*/M_\odot) > 10$), and only 15\% in lower-mass systems. In the massive regime, ISM absorption is seen in both star-forming and quiescent galaxies, whereas in lower-mass systems it is observed only in star-forming galaxies. In massive quiescent galaxies, Na D detectability appears linked to star formation history: it is preferentially detected in older systems with larger 4000 Å breaks, as well as younger, rapidly quenching galaxies with strong Balmer absorption $H\delta_A$. We identify Na D outflows in 26 galaxies, revealing a possible dichotomy in their driving mechanisms between star-forming and quiescent galaxies. In star-forming galaxies, outflow properties correlate with star-formation properties, consistent with a star-formation-driven origin. In quiescent galaxies, however, outflows are not associated with residual star formation and often require more energy than such star formation can provide. Together with the high AGN fraction among outflow-detected quiescent galaxies, this suggests that AGN dominate Na D-traced neutral outflows in cosmic noon quiescent systems. We further identify five quiescent galaxies with possible AGN fossil outflows, suggesting that AGN-driven outflows can persist beyond the active accretion phase and may help maintain quiescence.

Selected figures



2. NOEMA3D: Resolving radial gas flows in disk galaxies at $z \sim 1.1-1.6$ with high-resolution CO observations

Focus: galaxies

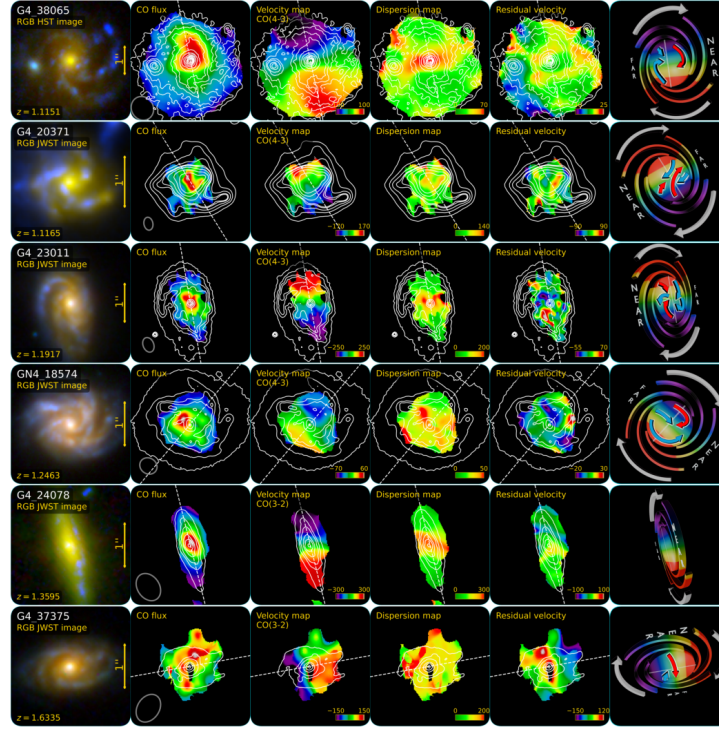
Authors: Jean-Baptiste Jolly, Linda J. Tacconi, Reinhard Genzel, Roberto Neri, Karl Schuster, Jianhang Chen, Natascha M. Förster Schreiber, Stavros Pastras, Letizia Scaloni, Giulia Tozzi, Capucine Barféty, Alberto Bolatto, Andreas Burkner, Françoise Combes, Pierre Cox, Ric Davies, Frank Eisenhauer, Juan Manuel Espejo Salcedo, Rodrigo Herrera-Camu, Santi García-Burillo, Tadayuki Kodama, Lilian Lee, Minju M. Lee, Daizhong Liu, Dieter Lutz, Giovanni Mazzolari, Thorsten Naab, Amit Nestor Shachar, Claudia Pulsoni, Alvio Renzini, Monica Rubio, Taro T. Shimizu, Amiel Sternberg, Eckhard Sturm, Hannah Übler, Antonio Usero, Stijn Wuyts

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PDF: <https://arxiv.org/pdf/2604.18503v1>

Summary: We present NOEMA3D, a unique high-resolution study of purely molecular gas kinematics at $z \sim 1.1$ to 1.6 , providing a dedicated view of cold gas dynamics at the late stages of the peak epoch of cosmic star formation. Using deep (*gtrsim* 20 hr on source per target) IRAM-NOEMA CO observations of 10 massive ($10.45 < \log(M^*/M_\odot) < 11.43$) main-sequence galaxies, complemented by high-resolution JWST imaging, we resolve the molecular gas kinematics and morphology on kiloparsec scales. We find that all galaxies exhibit ordered rotation with moderate intrinsic turbulence (median $\sigma_0 \sim 32 \pm 10$ km/s, median $V_c/\sigma_0 \sim 8.6 \pm 2.9$), consistent with dynamically turbulent disks at late cosmic noon. After modeling the axisymmetric rotation with the forward-modeling code *DysmalPy*, we reveal spatially coherent velocity residuals in all but one more inclined system. The inferred in-plane non circular motions reach amplitudes of ~ 50 - 100 km/s, significantly larger than typically observed in local disk galaxies. Interpreting these non-circular motions as radial flows we find that the velocity residuals spatially coincide with non-axisymmetric structures – spiral arms and bars – demonstrating a direct link between galaxy morphology and gas transport at $z \sim 1$ - 2 . In spiral galaxies, the residual velocity patterns are typically dominated by inflows, while barred systems display an apparent inflow-outflow pattern, characteristic of in-plane bar-driven gas motions. We further find that the inferred molecular gas inflow rates are substantial, with a typical net inflow rate of the order of the star formation rate ($\dot{M} \sim -50 M_\odot/\text{yr}$). This implies that spiral arms and bars at cosmic noon are highly efficient at funneling cold gas toward galaxy centers, perhaps driving the buildup of bulges and feeding central star forming regions and supermassive black holes.

Selected figures



3. NOEMA3D: Spatially resolved dust, CO, and [C I] in massive star-forming main sequence galaxies at cosmic noon

Focus: galaxies

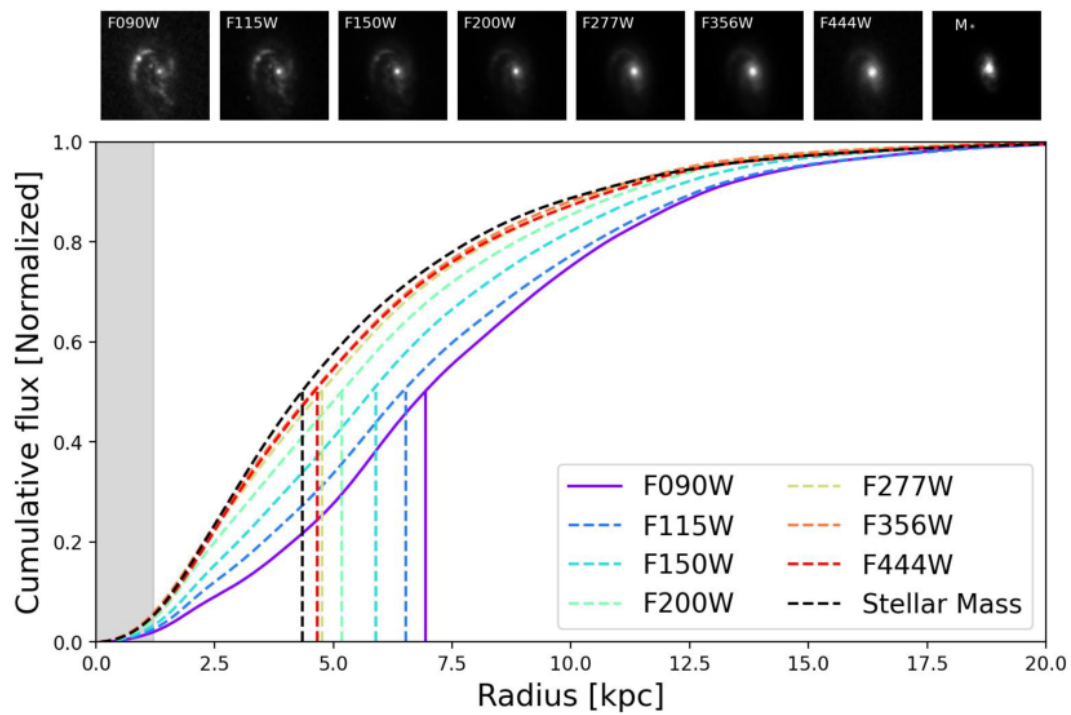
Authors: Jianhang Chen, Linda J. Tacconi, Reinhard Genzel, Roberto Neri, Karl Schuster, Natascha M. Förster Schreiber, Jean-Baptiste Jolly, Stavros Pastras, Letizia Scaloni, Giulia Tozzi, Capucine Barf  ty, Alberto Bolatto, Andreas Burkner, Fran  oise Combes, Pierre Cox, Ric Davies, Frank Eisenhauer, Juan Manuel Espejo Salcedo, Rodrigo Herrera-Camu, Santi Garc  a-Burillo, Tadayuki Kodama, Lilian Lee, Minju M. Lee, Daizhong Liu, Dieter Lutz, Giovanni Mazzolari, Thorsten Naab, Amit Nestor Shachar, Claudia Pulsoni, Alvio Renzini, Monica Rubio, Taro T. Shimizu, Amiel Sternberg, Eckhard Sturm, Hannah   bler, Antonio Usero, Stijn Wuyts

Published: 2026-04-20T16:53:44+00:00

PDF: <https://arxiv.org/pdf/2604.18504v1>

Summary: We present a spatially resolved study of cold molecular gas and dust in ten main-sequence galaxies at $z=1.1-1.6$, using observations of CO(4-3), CO(3-2), [C I](1-0), and dust continuum from the NOEMA3D survey. We find a widely presence of spatially extended molecular gas and dust, with sizes comparable to those of the stellar disk, in contrast to those of central-dominated starburst galaxies at similar redshifts. While various molecular gas tracers generally exhibit similar spatial distributions, the CO line (J=3-2 or J=4-3) remain the most effective for mapping molecular gas distribution and kinematics. In addition, the spatially resolved correlations between different molecular gas tracers exhibit about two times larger scatter than their galactic-integrated correlations, indicating that interstellar medium (ISM) conditions already deviate from global averages on scales of 3-6 kpc, likely reflecting the clumpy or inhomogeneous ISM in cosmic noon star-forming galaxies. Within our sample, both the molecular gas fraction and its depletion time are nearly constant across the galactic disks out to 2 Re, supporting a global linear Kennicutt-Schmidt law. The presence of extended molecular gas disks, along with regular stellar structures, small central bulges, and ordered cold gas kinematics, supports the idea that the evolution of main-sequence disk galaxies at cosmic noon is driven by steady gas accretion and transport through prominent spiral arms and/or bars. This process stands in contrast to the merger-driven stochastic gas accretion in compact starbursts.

Selected figures



4. Molecular Clouds at the Edge of the Galaxy II. Physical properties and scaling relations

Focus: galaxies

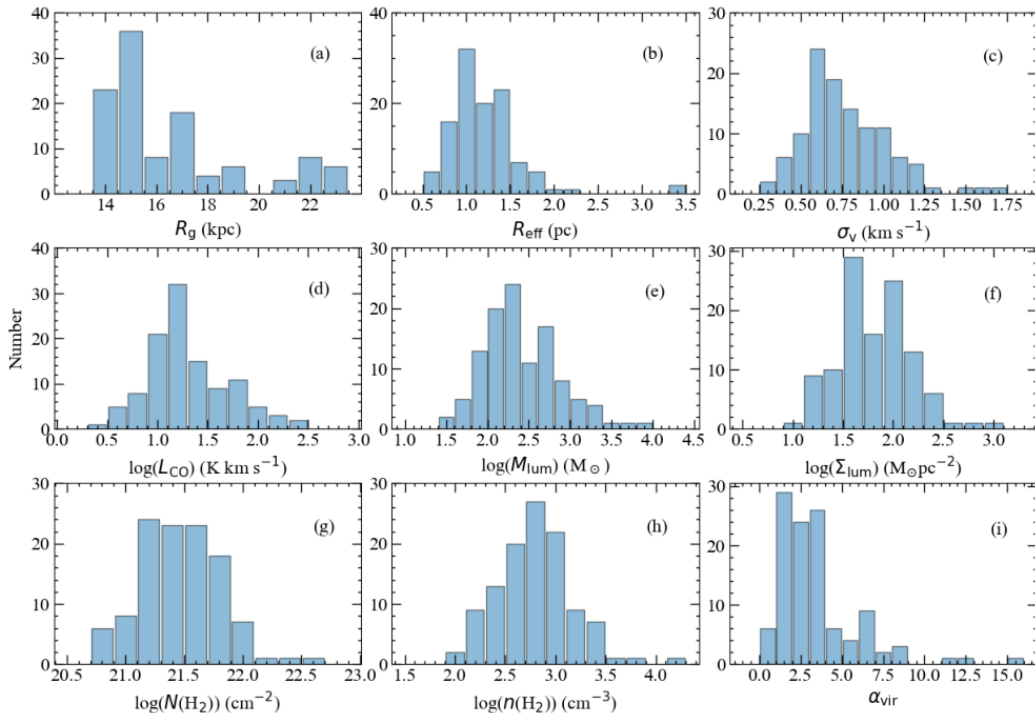
Authors: C. S. Luo, X. D. Tang, C. Henkel, Y. Sun, Y. Gong, X. W. Zheng, T. Liu, X. Lu, Y. P. Ao, X. P. Chen, D. L. Li, Y. X. He, K. Wang, J. W. Wu, J. Esimbek, J. J. Zhou, G. Wu, Y. X. Ma, W. A. Baan, J. J. Qiu, X. Zhao, J. S. Li, Q. Zhao, L. D. Liu, C. Y. Wang

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PDF: <https://arxiv.org/pdf/2604.18382v1>

Summary: The outer Galaxy presents an optimal setting for investigating molecular clouds and star formation in environments with low metallicity. A total of 72 Galactic edge clouds were surveyed using the CO(2–1) line with the IRAM 30m telescope, leading to the identification of 112 CO clumps within molecular clouds with linear resolutions of 0.5–0.9 pc. Parameters such as size, mass, surface density, and velocity dispersion of these CO clumps, derived from CO(2–1) observations, exhibit ranges of 0.6–3.4 pc, 34–8250 $M_\odot$, 12–1025 $M_\odot \text{pc}^{-2}$, and 0.3–1.7 km s^{-1} , respectively. Over the Galactocentric distance range of 14–23 kpc, no systematic variations are found in these parameters. The velocity dispersion-size relationship of the Galactic edge clumps is modeled as $\sigma_v = 0.69(\pm 0.03) R_{\text{eff}}^{0.36(\pm 0.10)}$, indicating that turbulence is present within the Galactic edge clumps, akin to observations in the inner Galactic disk clouds. Furthermore, the luminous mass-size relation of the Galactic edge clumps is described by $M_{\text{lum}} = 196(\pm 17) R_{\text{eff}}^{2.18(\pm 0.26)}$, suggesting the average column density remains almost constant for clouds of different sizes. The virial parameters range from 0.6 to 15.3, with a median value of 2.8 ± 0.6 , suggesting that most clumps are gravitationally unbound. Furthermore, the virial parameters of our Galactic edge clumps show a decreasing trend with increasing Galactocentric distances, described by an exponential relation $\alpha_{\text{vir}} = 33.0(\pm 10.4) e^{-R_g/6.7(\pm 0.9)}$, consistent with previous results.

Selected figures



5. Fourth-order galaxy-galaxy-lensing: Theoretical framework and direct estimation

Focus: galaxies, gravitational lensing

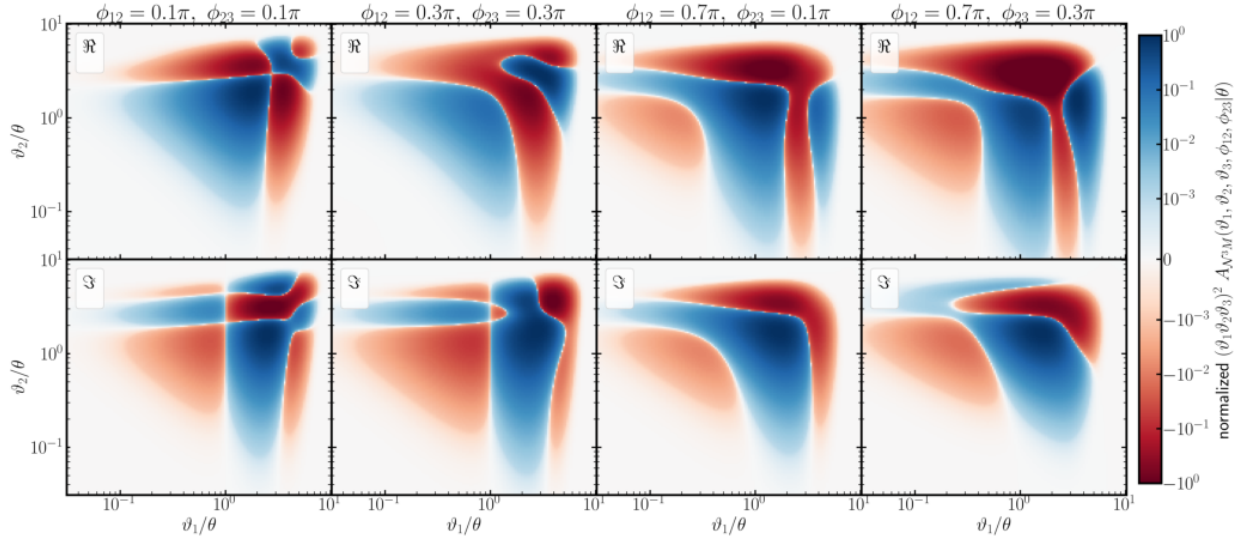
Authors: Jonathan Oel, Lucas Porth, Peter Schneider, Elena Silvestre-Rosello

Published: 2026-04-20T15:04:01+00:00

PDF: <https://arxiv.org/pdf/2604.18378v1>

Summary: Traditional galaxy-galaxy lensing is a well-established method of probing the statistical properties of the Universe’s matter and galaxy distribution. However, this measure does not carry all the statistical information, provided the matter and galaxy distribution contain non-Gaussian features. In order to study these non-Gaussianities, it is necessary to consider higher-order statistical measures. The aim of this work is to extend the analytical basis describing the statistical correlations between galaxies and shear to the fourth order, with special emphasis on the associated aperture statistics. In order to include fourth-order statistics in future analysis of the relation between mass and galaxies, we further investigate whether we can expect to detect these statistics from observations of stage IV surveys. We define the four-point correlation function (4PCF) between the shear and the positions of triplets of foreground galaxies and derive its relation to the respective trispectrum. We convert the 4PCF to aperture statistics and derive the analytical form of the respective filter function, which we then implement in a numerical integration pipeline. Furthermore, we develop a direct estimator that allows us to measure galaxy-mass aperture moments of arbitrary order on pixelized data using a Fast-Fourier-Transform (FFT) algorithm. We show that the corresponding aperture measure $\langle \mathcal{N}^3 M_{\text{ap}} \rangle$ can be calculated with sub-percent accuracy on relevant aperture scales, , by means of numerical integration. Furthermore, we apply the FFT-based direct estimator to a mock catalog with a realistic stage IV survey setup on a sky area of 2000 deg^2 , and detect the connected part of the aperture statistics $\langle \mathcal{N}^3 M_{\text{ap}} \rangle()$ with a signal-to-noise ratio of roughly nine on small aperture scales.

Selected figures



6. The MeerKAT Fornax Survey VII. Characterisation of the Fornax cluster’s magnetic field and new insights on magnetisation in large scale systems

Focus: galaxies

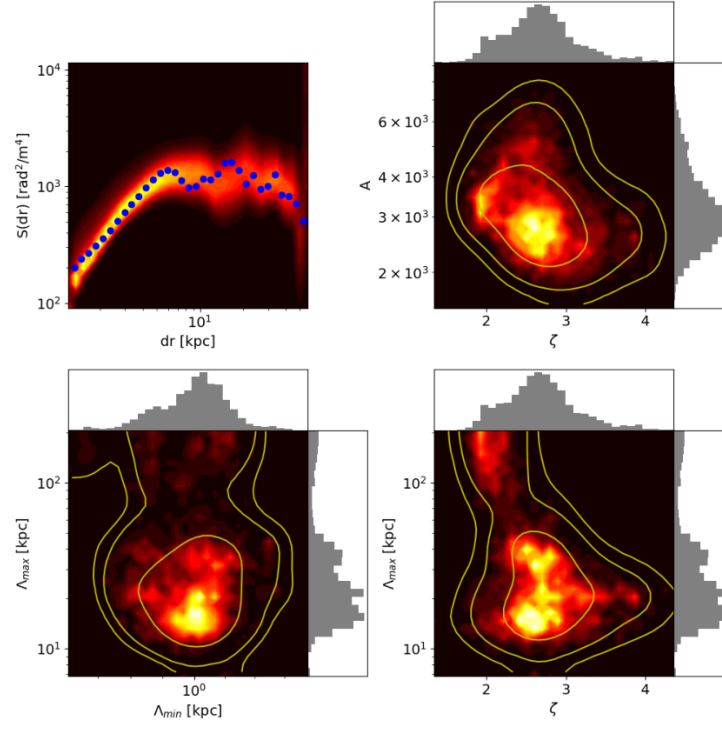
Authors: Francesca Loi, Matteo Murgia, Federica Govoni, Paolo Serra, Valentina Vacca, Filippo Marcello Maccagni, Peter Kamphuis, Dane Kleiner

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PDF: <https://arxiv.org/pdf/2604.18338v1>

Summary: Large scale magnetic fields in galaxy clusters can influence their physics and the evolution of the cluster galaxies. These properties remain poorly constrained due to a historical lack of high-sensitivity and high-resolution spectro-polarimetric data. Thanks to the advent of the SKA pathfinders and precursors this is now dramatically changing. By exploiting the densest RM grid produced to date with broadband spectro-polarimetric data in the context of the MeerKAT Fornax Survey and presented in a previous paper (508 sources over 6.35 deg^2), we aim to study in detail the Fornax cluster’s magnetic field. We compare the RM grid properties with numerical simulations to constrain the strength and the structure of the intra-cluster magnetic field. We model the magnetic field power spectrum with a power-law and we find a slope of ~ 2.7 fluctuating between ~ 1 and $\sim 15 \text{ kpc}$. It has a central strength of $\sim 5 \mu\text{G}$ decreasing with the thermal plasma density according to a power-law exponent of ~ 1.6 , the highest value to date in large scale systems. By analysing a sample of 17 galaxy clusters and groups with magnetic field estimates from the literature, we observe larger auto-correlation lengths in the case of massive merging clusters and lower values for relaxed clusters and low-mass systems. We also observe a systematic increase of the central magnetic field strength as a function of central density, $B_0 \propto n_0^{0.38}$. Finally, we argue that the steepening of the Fornax cluster’s magnetic field profile and its relatively high central strength could be indicative of a recent re-amplification at the centre due to the extended central radio galaxy. The sample analysis supports the proposed scenario, although more detailed magnetic field studies conducted using consistent modelling on larger samples are needed to better understand magnetisation in clusters and groups.

Selected figures



7. Characterizing the velocity anisotropy of the Milky Way’s stellar halo

Focus: galaxies

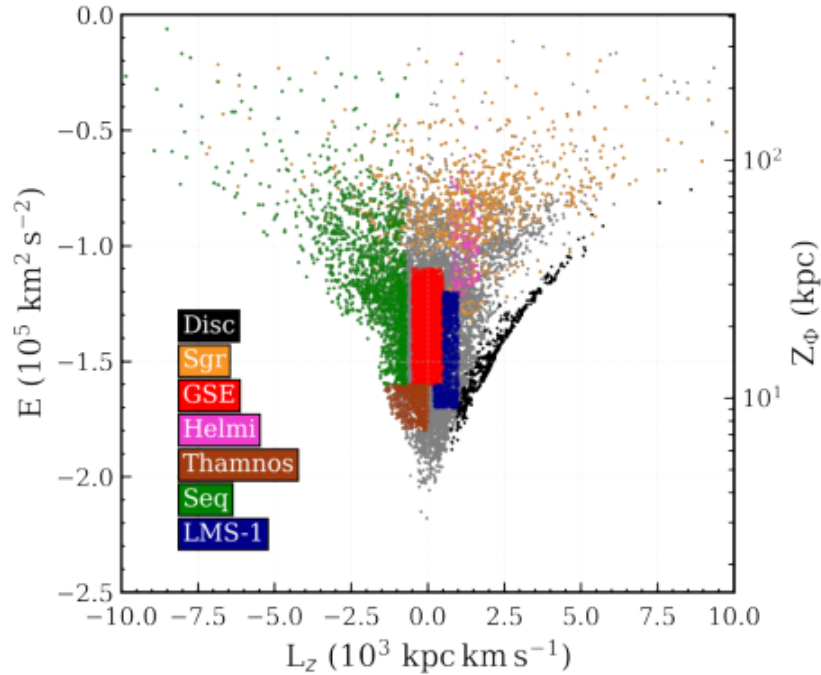
Authors: F. O. Barbosa, A. Pérez-Villegas, S. Rossi, R. M. Santucci, L. Aguilar, H. D. Perotoni, G. Limberg, L. Borbolato, J. V. Nogueira-Santos

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PDF: <https://arxiv.org/pdf/2604.18278v1>

Summary: Modeling the Milky Way stellar halo requires well-determined density and velocity anisotropy profiles. However, it has been challenging to gather a large sample of stars with six-dimensional data that extend beyond 40 kpc to map the outer halo. Our work investigates the velocity anisotropy in the Milky Way stellar halo with more than 10,000 blue horizontal-branch stars, combining Gaia astrometric data and spectroscopic data from SEGUE, DESI and LAMOST. This large sample allows us to obtain a detailed profile of up to ~ 70 kpc. Radial velocities are predominant in the inner halo (< 30 kpc), and the anisotropy presents a smooth decrease before rapidly dropping to negative values, becoming dominated by tangential dispersion velocities. Removing the main known accreted structures of the Milky Way, makes the anisotropy profile radially-dominated at all radii. Our profile clearly shows an increase in the anisotropy from the center of the Galaxy, in accordance to the simulations. We also investigate the correlation of anisotropy with metallicity and with color. The lack of correlation between metallicity and anisotropy in our clean sample reinforces that this relation is driven by merger events. The initial exploration with color indicates a relation between kinematics and age, showing that older stars are dynamically colder and present less radial orbits than younger stars in the inner halo.

Selected figures



8. From Gaia to GaiaNIR: I. Probing dark matter halos in globular clusters

Focus: dark matter

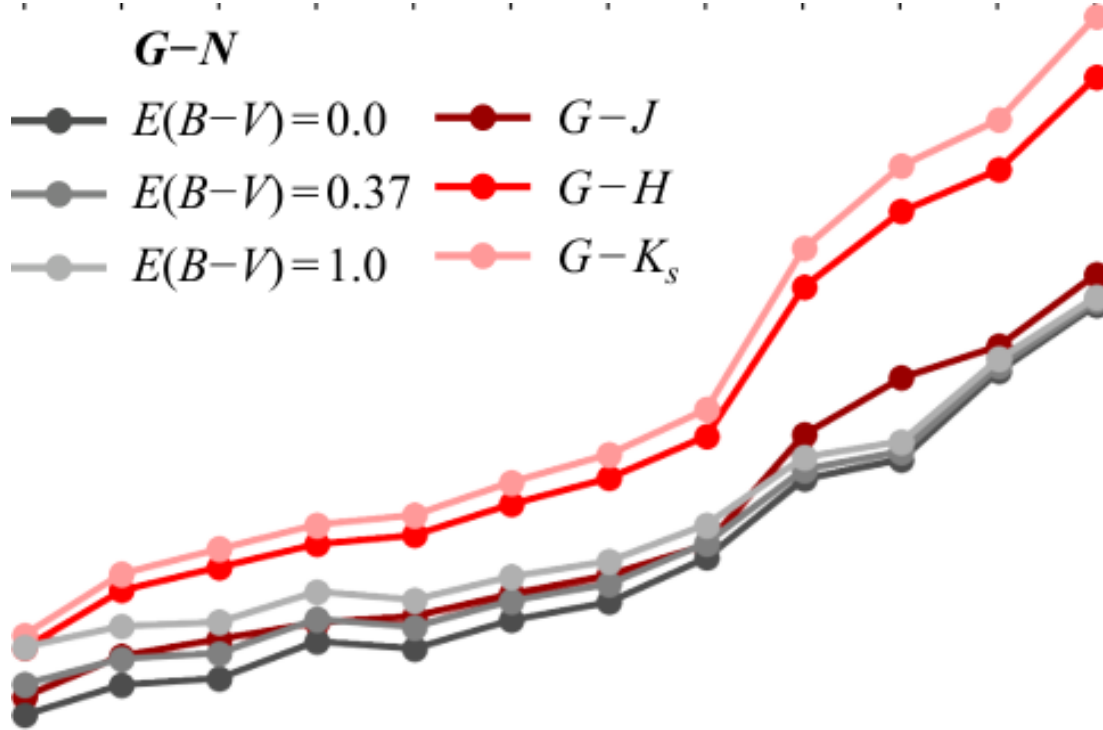
Authors: I. Henum, D. Hobbs, Ó. Jiménez-Arranz, P. J. McMillan, R. P. Church

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PDF: <https://arxiv.org/pdf/2604.18265v1>

Summary: The proposed GaiaNIR mission would extend Gaia’s astrometric capabilities into the near-infrared, improving astrometric precision and enabling observations in heavily dust-obscured regions. In this work, we investigate the impact of GaiaNIR on the detectability of dark matter halos in globular clusters by comparing its performance with that of Gaia. Expected observations from future Gaia data releases and GaiaNIR are modeled for a globular cluster with properties similar to M4. The cluster is simulated with a range of dark matter halo sizes and varying levels of extinction, allowing a direct assessment of each mission’s ability to detect and distinguish dark matter halos across different extinction conditions. To support this comparison, the relationship between the Gaia G band and several near-infrared bands is examined. We find that Gaia can resolve extended dark matter halos in low-extinction conditions, but its performance degrades significantly as extinction increases. In contrast, GaiaNIR reduces statistical uncertainties and retains sensitivity to extended halos even with strong extinction. These results indicate that GaiaNIR would both strengthen constraints on kinematic signatures accessible to Gaia and enable the study of clusters in heavily obscured regions that are currently beyond Gaia’s reach.

Selected figures



9. Galaxy Populations in the IllustrisTNG Caustic Skeleton

Focus: galaxies, dark matter

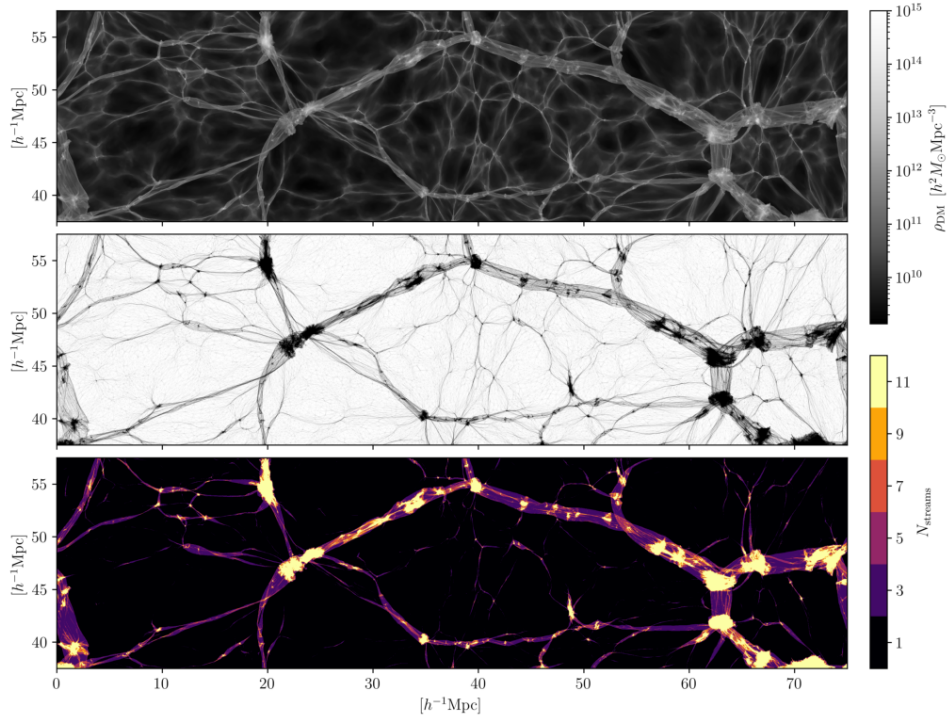
Authors: Benjamin Hertzsch, Job Feldbrugge, Rien van de Weygaert

Published: 2026-04-20T12:56:25+00:00

PDF: <https://arxiv.org/pdf/2604.18209v1>

Summary: The caustic skeleton is a parameter-free and mathematically rigorous formalism for tracing the hierarchical formation history of the multiscale cosmic web from the singularities in the underlying dark matter flow. In the present study, we explicitly use the multistreaming nature of the cosmic mass distribution to address the influence of the weblike embedding on the galaxy populations and discern their properties in different web environments. To this end, we construct the multiscale caustic skeleton of the dark mass distribution in the state-of-the-art suite of the large-scale IllustrisTNG simulations. In addition to the multistreaming dark matter density field, we assess the characteristic properties of the intergalactic baryonic gas in the vicinity of the caustics. Next, we associate the galaxies with the voids, walls, filaments and cluster nodes, and investigate their colours and star formation activities. A unique feature of the analysis is that it explicitly addresses the multiscale aspects with respect to the galaxy population, assessing issues such as the fraction of (blue) galaxies as a function of the scale of the cosmic web pattern and its caustic features. We find that the galaxy properties form a continuum in the scale-space cosmic web. Intimately coupled to the hierarchical build-up of the cosmic structure, it also allows us to systematically assess the impact of the formation time of the various structural components of the cosmic web on the galaxy properties. This furthers insight into the establishment of the observed colour-density relation of galaxies.

Selected figures



10. Latest results from the IceCube Neutrino Observatory

Focus: dark matter

Authors: Thijs Juan van Eeden

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PDF: <https://arxiv.org/pdf/2604.18178v1>

Summary: The IceCube Neutrino Observatory has opened a new window into the high-energy Universe, providing measurements of neutrinos over a broad energy range. This contribution presents recent results, including a follow-up on the first identification of a steady neutrino source NGC 1068, measurements of the flavor composition of the diffuse astrophysical flux, limits on prompt atmospheric neutrinos, and searches for neutrinos from dark matter annihilation in the Sun. These measurements probe neutrino production mechanisms, fundamental particle interactions, and physics beyond the Standard Model. Looking forward, the recently deployed IceCube Upgrade will enhance sensitivity to lower-energy neutrinos and reduce systematic uncertainties, while the planned IceCube-Gen2 will expand the detector volume, increase the neutrino detection rate, and extend energy reach, enabling more detailed studies of cosmic sources and high-energy particle physics.

Selected figures

